Lu Yi

LLM Agents, Recommendation, Graph Neural Networks
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Education

Renmin University of China, Beijing, China 2022 – Present

Phd Candidate in Artificial Intelligence (AI), GPA 3.88/4.0

- Supervisor: Prof. *Zhewei Wei* at the *ALGO Lab*
- Research Interests: LLM Agents, Recommendation, Graph Neural Networks
- Awards: **China National Scholarship**

Beijing University of Posts and Telecommunications, Beijing, China 2018 – 2022

B.S. in Computer Science (CS), GPA 91.24/100, Rank 2.5%

- **Awards:** Outstanding Graduate, First Prize Scholarship, Jiongpan Zhou Scholarship (Only one student in the school is selected each year), First Prize in NSCSCC 2020

Internships

Alibaba: Tongyi Group 2025.05 – Present

Research Intern | Mentors: Liuyi Yao, Yuexiang Xie, Yaliang Li

Open-source Contributions:

- AgentScope: Designed a hierarchical memory system for context and long-term memory management.
  ❶ Automatic long-context processing: supports unbounded context windows via dynamic compression.
  ❷ Context offloading: persists non-critical content to the file system and retrieves on demand.
  ❸ Vector database integration: enables semantic retrieval and reasoning over long-term memory.
  ❹ Achieves 100%+ relative gain in mean@3 on BrowseComp-Plus with Qwen3-max.
- Trinity-RFT: Resolved over-rollout inefficiency by early-stopping incomplete groups once a target completion ratio is met, leveraging all available completed rollouts for training.

Agentic Context Manager: Fine-tunes Qwen3-4B via agentic RL to autonomously manage context during task execution, addressing context rot in long-horizon agent tasks.

- Rollout filtering for multi-step GRPO: discards slow rollouts (typically from failed agent runs) to accelerate training and improve reward signal quality.
- Process reward design: evaluates context manager quality based on subsequent agent behavior—better agent actions indicate effective context pruning.
- Achieves 100%+ relative gain in mean@3 on BrowseComp-Plus with Qwen3-max, GPT-4o-mini, Gemini-2.5-flash, and Seed-1.6-flash, etc.

Publications

Future Link Prediction Without Memory or Aggregation

Lu Yi, *Runlin Lei, Fengran Mo, Yanping Zheng, Zhewei Wei, Yuhang Ye. NeurIPS, 2025. [PDF]*

A simple yet effective method for future link prediction, excel in real-world recommendations.

TGB-Seq Benchmark: Challenging Temporal GNNs with Complex Sequential Dynamics

Lu Yi, *Jie Peng, Yanping Zheng, Fengran Mo, Zhewei Wei, et al. ICLR 2025. [PDF] [Website]*

A new benchmark for future link prediction involving complex sequential dynamics.

Scalable and Certifiable Graph Unlearning: Overcoming the Approximation Error Barrier

Lu Yi, *Zhewei Wei. ICLR 2025 (**Spotlight**). [PDF]*

The first approach to scale certified graph unlearning to billion-edge graphs.

Optimal Dynamic Subset Sampling: Theory and Applications

***Lu Yi**, Hanzhi Wang, Zhewei Wei. KDD 2023 (**Oral**). [PDF]*

The first dynamic subset sampling algorithm with optimal sampling and update time complexity.

Random-Walk Probability Estimation on Dynamic Weighted Graphs

Hanzhi Wang, **Lu Yi***, Zhewei Wei. Journal of Computer Research and Development (J-CRAD), 2024. Invited **oral** presentation in China Conference on Data Mining (CCDM), 2024. [PDF]*

A random walk framework by coin-flip sampling to estimate L-hop random-walk probabilities.

Robustness in Text-Attributed Graph Learning: Insights, Trade-offs, and New Defenses

*Runlin Lei, **Lu Yi**, Mingguo He, Pengyu Qiu, Zhewei Wei, et al. ICLR 2026. [PDF]*

Systematic evaluation of TAG model robustness against adversarial attacks.

A survey of dynamic graph neural networks

*Yanping Zheng, **Lu Yi**, Zhewei Wei. Frontiers of Computer Science, 2024. [PDF]*

A comprehensive review of dynamic graph neural networks.

♡ Personal AI Tools

CCManager: Multi-Device Task Manager for Claude Code *[CODE]*

- Orchestrate Claude Code tasks across multiple devices via mobile-friendly Web UI with voice input.
- Security-first design: Docker sandbox isolation, Git worktree for safe parallel development.
- Real-time streaming output, session resume, and plan-mode review for human-in-the-loop control.